

F-Test and Analysis of Variance

8.1 F-DISTRIBUTION

The Fisher's *F-distribution* is defined as a distribution of the ratio of two independent chi-square variables each divided by the corresponding degrees of freedom. The probability density function of F-distribution is given by

$$p(F) = y_0 \frac{F^{(v_1-2)/2}}{\left(1 + \frac{v_1}{v_2} F\right)^{(v_1+v_2)/2}} ; 0 \leq F < \infty \quad (8-1)$$

where $y_0 = a \text{ constant}$ and, $v_1, v_2 = \text{degrees of freedom}$

$$F = \frac{\chi_1^2/v_1}{\chi_2^2/v_2}, \quad \text{when } v_1 > v_2 \quad (8-2)$$

Like other statistics t, χ^2 etc., if the null hypothesis is true, then F statistic has a particular sampling distribution as in **Fig. 8-1**.

F-distribution is actually a whole family of distributions each of which is identified by a pair of degrees of freedom. Referring to **Fig. 8-1**, the first number within the small braces represent the number of degrees of freedom in the numerator of *F-ratio*, the second, the degrees of freedom in the denominator. It is obvious that F-distribution has a single mode. Note that the shape of F-distribution depends on the value of degrees of freedom and the value of F lies between 0 to ∞ (zero to infinity). However, F-distribution is generally skewed to the right and tends to be more symmetrical as the number of degrees of freedom (v_1, v_2) increase.

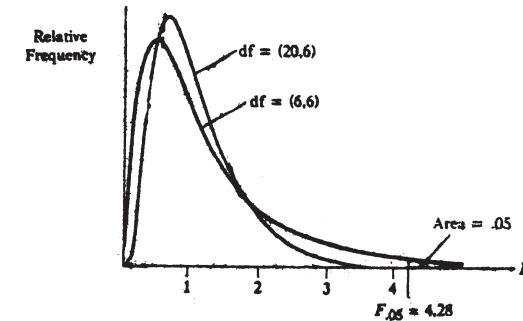


Fig. 8-1 : The F-distributions for various degrees of freedom

8.2 F-TEST OR VARIANCE RATIO TEST

The *F-test*, also called *variance ratio test*, is based on *F-distribution*. The steps used in testing the significance of two sample estimates of a population variance (or equality of population variances) are as follows :

Step 1. *Formulate the null and alternative hypotheses.*

$$H_0 : \sigma_1^2 = \sigma_2^2 = \dots = \sigma^2$$

i.e. the independent estimates of the common population variance do not differ significantly.

$$H_1 : \sigma_1^2 \neq \sigma_2^2 \neq \dots \neq \sigma^2$$

Step 2. *Under H_0 , compute the test statistic,*

$$F = \frac{S_1^2}{S_2^2} \quad \text{if } S_1 > S_2 \quad (8-3)$$

$$F = \frac{S_2^2}{S_1^2} \quad \text{if } S_2 > S_1 \quad (8-4)$$

$$\text{where } S_1^2 = \frac{1}{n_1 - 1} \sum (X_1 - \bar{X}_1)^2 = \frac{n_1}{n_1 - 1} s_1^2$$

$$S_2^2 = \frac{1}{n_2 - 1} \sum (X_2 - \bar{X}_2)^2 = \frac{n_2}{n_2 - 1} s_2^2$$

$$s_1^2 = \frac{1}{n_1} \sum (X_1 - \bar{X}_1)^2 = \text{actual sample variance of } X_1$$

$$s_2^2 = \frac{1}{n_2} \sum (X_2 - \bar{X}_2)^2 = \text{actual sample variance of } X_2$$

Step3. Write down the tabulated value of F for $\{(n_1 - 1), (n_2 - 1)\}$ degrees of freedom at $\alpha = 5\%$ level of significance unless otherwise stated from Appendix 4.

Step4. Make decision by applying the following decision rules :

- (i) If the computed value of F is less than its tabulated value, H_0 is accepted.
- (ii) If the computed value of F is greater than its tabulated value, H_0 is rejected i.e. H_1 is accepted.

Example 1.

Two random samples drawn from two normal populations

Sample I 17 27 18 25 27 29 27 23

Sample II 16 16 20 16 20 17 15 21

Using F-test whether the two samples have come from the same population.

Solution :

H_0 : $\sigma_1^2 = \sigma_2^2$ i.e. the samples are from the same population.

H_1 : $\sigma_1^2 \neq \sigma_2^2$ i.e. the samples are not from the same population.

Computation of test statistic : $F = \frac{S_1^2}{S_2^2}$

Computation of sample variances

Sample I			Sample II		
X_1	$d_1 = X_1 - 25$	d_1^2	X_2	$d_2 = X_2 - 16$	d_2^2
17	-8	64	16	0	0
27	2	4	16	0	0
18	-7	49	20	4	16
25	0	0	16	0	0
27	2	4	20	4	16
29	4	16	17	1	1
27	2	4	15	-1	1
23	-2	4	21	5	25
$\sum d_1 = -7$		$\sum d_1^2 = 145$	$\sum d_2 = 13$		$\sum d_2^2 = 59$

$$\sum (X_1 - \bar{X}_1)^2 = \sum d_1^2 - \frac{(\sum d_1)^2}{n_1} = 145 - \frac{(-7)^2}{8}$$

$$= 145 - 6.125 = 138.875$$

$$\sum (X_2 - \bar{X}_2)^2 = \sum d_2^2 - \frac{(\sum d_2)^2}{n_2} = 59 - \frac{(13)^2}{8}$$

$$= 59 - 21.125 = 37.875$$

$$S_1^2 = \frac{1}{n_1 - 1} \sum (X_1 - \bar{X}_1)^2 = \frac{1}{8 - 1} \times 138.875 = 19.84$$

$$S_2^2 = \frac{1}{n_2 - 1} \sum (X_2 - \bar{X}_2)^2 = \frac{1}{8 - 1} \times 37.875 = 5.41$$

Since $S_1^2 > S_2^2$, the test statistic is : $F = \frac{S_1^2}{S_2^2} = \frac{19.84}{5.41} = 3.67$

d.f. for large estimate $v_1 = 8 - 1 = 7$; d.f. for smaller estimate
 $= v_2 = 8 - 1 = 7$

Table value of F at 5% level of significance for $v_1 = 7$ and $v_2 = 7$ is 3.79.

Decision : Since the computed value of F is less than its tabulated value, H_0 is accepted i.e. the two samples have come from the same population.

Example 2.

Two independent samples of 7 and 6 items respectively had the following values of variables.

Weights in lbs.

Sample A 9 11 13 11 15 9 12

Sample B 10 12 10 14 9 8

Are the two estimates of population variance differ significantly ?

(For $v_1 = 6$, $v_2 = 5$, $F_{0.05} = 4.95$, For $v_1 = 5$, $v_2 = 6$, $F_{0.05} = 4.39$)

Solution :

H_0 : $\sigma_1^2 = \sigma_2^2$ i.e. there is no significant difference between the variances estimated by the two samples A and B.

H_1 : $\sigma_1^2 \neq \sigma_2^2$

Computation of sample variances for finding F

Sample A			Sample B		
X_1	$d_1 = X_1 - 11$	d_1^2	X_2	$d_2 = X_2 - 10$	d_2^2
9	-2	4	10	0	0
11	0	0	12	2	4
13	2	4	10	0	0
11	0	0	14	4	16
15	4	16	9	-1	1
9	-2	4	8	-2	4
12	1	1			
$\sum d_1 = -3$ $\sum d_1^2 = 29$			$\sum d_2 = 3$ $\sum d_2^2 = 25$		

$$\sum (X_1 - \bar{X}_1)^2 = \sum d_1^2 - \frac{(\sum d_1)^2}{n_1} = 29 - \frac{(-3)^2}{7} = 29 - 1.3 = 27.7$$

$$\sum (X_2 - \bar{X}_2)^2 = \sum d_2^2 - \frac{(\sum d_2)^2}{n_2} = 25 - \frac{(3)^2}{6} = 25 - 1.5 = 23.5$$

$$S_1^2 = \frac{1}{n_1 - 1} \sum (X_1 - \bar{X}_1)^2 = \frac{1}{7 - 1} \times 27.7 = 4.62$$

$$S_2^2 = \frac{1}{n_2 - 1} \sum (X_2 - \bar{X}_2)^2 = \frac{1}{6 - 1} \times 23.5 = 4.70$$

Since $S_1^2 < S_2^2$, the test statistic is : $F = \frac{S_2^2}{S_1^2} = \frac{4.70}{4.62} = 1.02$

Table value of F at 0.05 level of significance for $v_1 = 6$ and $v_2 = 5$ is 4.39

Decision : Since $F < F_{0.05} (5, 6)$ H_0 is accepted i.e. there is no significant difference between the variance estimated by the two samples A and B.

Example 3.

Two random samples drawn from normal population are :

Sample I 20 16 26 27 23 22 18 24 25 19

Sample II 17 23 32 25 22 24 28 18 31 33 20 27

Obtain estimates of the populations variances and test whether the two population have the same variance. Use F-test.

Solution :

$H_0 : \sigma_1^2 = \sigma_2^2$ i.e. the two population have the same variance

$H_1 : \sigma_1^2 \neq \sigma_2^2$ i.e. the two population have not the same variance.

Computation of sample variances for finding F

Samples I			Sample II		
X_1	$x_1 = X_1 - X_1$	x_1^2	X_2	$x_2 = X_2 - X_2$	x_2^2
20	-2	4	17	-8	64
16	-6	36	23	-2	4
26	4	16	32	7	49
27	5	25	25	0	0
23	1	1	22	-3	9
22	0	0	24	-1	1
18	-4	16	28	3	9
24	2	4	18	-7	49
25	3	9	31	6	36
19	-3	9	33	8	64
			20	-5	25
			27	2	4
$\sum X_1 = 220$ $\sum X_1^2 = 120$			$\sum X_2 = 300$ $\sum X_2^2 = 314$		

$$\bar{X}_1 = \frac{\sum X_1}{n_1} = \frac{220}{10} = 22 \quad \bar{X}_2 = \frac{\sum X_2}{n_2} = \frac{300}{12} = 25$$

$$S_1^2 = \frac{\sum x_1^2}{n_1 - 1} = \frac{120}{10 - 1} = 13.33, S_2^2 = \frac{\sum x_2^2}{n_2 - 1} = \frac{314}{12 - 1} = 28.55$$

Since $S_1^2 < S_2^2$, Under H_0 , the test statistic is : $F = \frac{S_2^2}{S_1^2}$

$$= \frac{28.55}{13.33} = 2.14$$

For larger estimate, $v_2 = 12 - 1 = 11$; For smaller estimate,

$$v_1 = 10 - 1 = 9$$

Table value of F at 0.05 level for $v_2 = 11$ and $v_1 = 9$ is 3.175

Decision : Since the computed value of F is less than its tabulated value, null hypothesis is accepted i.e. the two population have the same variance.

8.3 ANALYSIS OF VARIANCE

Analysis of variance (ANOVA) was developed by R.A. Fisher. *F-test* (or *Fisher's test*) is widely used in the analysis of variance. It is basically used to test the hypothesis of equality between two variances. F-test is also used to test the hypothesis of equality amongst several means. F-test is particularly suitable for experimental work where the assumption of equality of variances is not necessary. The assumptions made in ANOVA are:

- (i) The population for each sample must be normally distributed with the identical mean and variance except for large sample sizes.
- (ii) All the sample observations must be randomly selected and independent.
- (iii) $\frac{\sigma_1^2}{\sigma_2^2} \geq 1$ i.e. Variance ratio must be greater than 1
- (iv) $0 \leq F \leq \infty$ i.e. the value of F must lie between zero to infinity.

ANOVA is mainly carried out under :

- (i) one-way classification
- (ii) two-way classification.

The actual ANOVA is performed on the basis of a ratio between variances. F-statistic is the ratio obtained by dividing the variance between samples by the variances within samples and is given by

$$F = \frac{\text{Variance between samples}}{\text{Variance within samples}} \quad (9-5)$$

In general, the variance between samples is greater than the variance within samples. Sometimes if the variance within samples happens to be greater than the variance between samples, then the variances should be interchanged so that F is always greater than 1.

8.3.1 ANOVA in One Way Classification

In one way classification, the effect of any one factor is taken into consideration. For example, the effect of the use of one or more types of fertilisers may be considered on several pieces of land. The

different types of fertilisers are, generally represented by the columns and, the yields of sample plots under each type of fertiliser are placed in the respective columns. ANOVA in one-way classification can be carried out by three methods :

1. Direct Method 2. Short-cut Method 3. Coding Method

8.3.1.1 Direct Method

Step 1. *Formulate null and alternative hypothesis.*

$$H_0 : \mu_1 = \mu_2 = \dots \dots \mu_k$$

i.e. the arithmetic means of population from which k samples are drawn are equal to one another.

$$H_1 : \mu_1 \neq \mu_2 \neq \dots \dots \neq \mu_k$$

Step 2. *Compute the test statistic, F.* For this we use the following procedures.

(a) Compute the mean of each column i.e. $\bar{X}_1, \bar{X}_2, \dots, \bar{X}_k$.

(b) Find the mean of the samples mean or grand mean

$$M = \bar{X} = \frac{\bar{X}_1 + \bar{X}_2 + \dots + \bar{X}_k}{n_1 + n_2 + \dots + n_k}$$

(c) Find the deviation of the sample means from the grand mean, square these deviations and multiply by sample size and find the sum of these figures. This is called *SSC or SSB (Sum of squares in columns or between samples)*

(d) Compute *mean sum of square of columns (MSC)* by using the relation

$$MSC = \frac{SSC}{k-1}$$

where k = the number of samples

$k-1$ = degrees of freedom

(e) Compute *SSE or SSW (sum of square of variation within samples)* by using the following procedures.

- (i) Find the mean of the sample

- (ii) Sum the squares of deviation of various sample items from their mean.
- (iii) Repeat (ii) for all samples, and obtain (*SSE*) the *total of sum of squares of the deviation of various samples from their respective means*.
- (f) Calculate mean sum of squares of error by using.

$$MSE = \frac{SSE}{N-k} \text{ where } N = \text{Total number of observations}$$

- (g) Calculate the total sum of squares of variations by using $SST = SSC + SSE$.

- (h) Prepare ANOVA table as follows :

Sources of variation	Sum of squares	Degrees of freedom	Mean sum of squares	F - ratio
Between samples	SSC	$k - 1$	$MSC = \frac{SSC}{k - 1}$	$F = \frac{MSC}{MSE}$
Within samples	SSE	$N - k$	$MSE = \frac{SSE}{N - k}$	
Total	SST	$N - 1$		

It is important to note that while computing *F-ratio*, if variance within the sample happens to be more than variance between the samples, the numerator and denominator should be interchanged and degrees of freedom adjusted accordingly.

Step 3. Write down the critical value of *F* for $(k - 1, N - k)$ degrees of freedom at $\alpha = 5\%$ level of significance unless otherwise stated.

Step 4. Make decision. If the computed value of *F* is less than its critical value, H_0 is accepted otherwise H_1 is accepted.

The above steps will be more clear from the solved examples.

Example 4.

The following are the yields in quintals per hectare of a certain variety of vegetable crop grown in a particular type of soil treated with three chemical fertilizers A, B and C.

A	B	C
3	2	4
4	4	6
5	3	5
4	3	5

Set up a table of analysis of variance by using direct method. State whether the difference between the yields of the three treatments is significant. Given $F = 4.26$ at 5% level with $\nu_1 = 2$ and $\nu_2 = 9$.

Solution :

Null hypothesis (H_0) : $\mu_1 = \mu_2 = \mu_3$ i.e. there is no difference between the treatments with regard to yields.

Alternative hypothesis (H_1) : $\mu_1 \neq \mu_2 \neq \mu_3$ i.e. the treatments are distinct with regard to yields.

Computations for Grand Mean and Sums of Squares

Let $x_1 = X_1 - \bar{X}_1$, $x_2 = X_2 - \bar{X}_2$ and $x_3 = X_3 - \bar{X}_3$

X_1	X_2	X_3	x_1^2	x_2^2	x_3^2	$(\bar{X}_1 - M)^2$	$(\bar{X}_2 - M)^2$	$(\bar{X}_3 - M)^2$
3	2	4	1	1	1	0	1	1
4	4	6	0	1	1	0	1	1
5	3	5	1	0	0	0	1	1
4	3	5	0	0	0	0	1	1
16	12	20	2	2	2	0	4	4
$\bar{X}_1 = 4$	$\bar{X}_2 = 3$	$\bar{X}_3 = 5$						

$$\text{Grand Mean } M = \frac{\bar{X}_1 + \bar{X}_2 + \bar{X}_3}{N} = \frac{4 + 3 + 5}{3} = 4$$

Sum of the squares between the samples

$$\begin{aligned} SSC &= n_1 (\bar{X}_1 - M)^2 + n_2 (\bar{X}_2 - M)^2 + n_3 (\bar{X}_3 - M)^2 \\ &= 4 (4 - 4)^2 + 4 (3 - 4)^2 + 4 (5 - 4)^2 \\ &= 0 + 4 + 4 = 8 \end{aligned}$$

Sum of the squares within the samples (SSE)

$$= \sum (X_1 - \bar{X}_1)^2 + \sum (X_2 - \bar{X}_2)^2 + \sum (X_3 - \bar{X}_3)^2$$

$$= 2 + 2 + 2 = 6$$

ANOVA Table

Sources of variation	Sum of squares (S.S.)	Degrees of freedom (d.f.)	Mean sum of squares (MS)	Variance ratio (F)
Between treatment	8	$3 - 1 = 2$	$8/2 = 4$	$F = 4/0.67 = 5.97$
Within treatment	6	$12 - 3 = 9$	$6/9 = 0.67$	
Total	14	$12 - 1 = 11$		

Table value of F for $v_1 = 2$ and $v_2 = 9$ at 5% level of significance is 4.26

Decision : Since the computed value of F is greater than its tabulated value, H_0 is rejected i.e. there is significant difference between the treatments with regard to yields at 5% level. We conclude that the treatments are distinct at 5% level with regard to yields.

8.3.1.2 Short cut method

Direct method of computing F-ratio is tedious and time consuming. So it is desirable to use short-cut method and its steps are as follows :

- Find the sum of the values of all the items of all the samples and denote it by T .
- Calculate the *correction factor* $= T^2/N$ where N = total number of items in all the samples.
- Find the squares of all the items and add them together.
- Find out the total sum of squares (SST) by subtracting $\frac{T^2}{N}$ from the sum of squares of all the items of the samples.

- Find out the sum of squares between the samples (SSC) by subtracting T^2/N from the sum of the squares of the totals divided by the number of items in each sample
- Find the sum of squares within sample by using $SSE = SST - SSC$.
- Prepare ANOVA table to compute F .

Sources of variation	Sum of squares	Degrees of freedom	Mean sum of squares	F - ratio
Between samples	SSC	$k - 1$	MSC	$\frac{MSC}{MSE}$
Within samples	SSE $= SST - SSC$	$N - k$ $= (N - 1) - (k - 1)$	$= \frac{SSC}{k - 1}$ $MSE = \frac{SSE}{N - k}$	
Total	SST	$N - 1$		

Rest steps 1, 3, and 4 are the same as those of Direct Method

8.3.1.3 Coding Method

The value of variance ratio (F) does not change if all the observations are multiplied or divided by a common factor or if a common factor is added to or subtracted from each observation. This property can be used to compute F -ratio. When the given observations are large we reduce them to a simplified set of figures by coding them. Once the given observations are reduced to a convenient form, then all the steps of the short-cut method are used for the computation and interpretation of the F -ratio. The steps 1, 3 and 4 are the same as those of Direct Method.

Example 5.

A teacher wishes to test three different teaching methods, I, II, III. To do this, three groups of 5 students each are chosen at random, and each group is taught by a different method. The same examination is then given to all the students and the marks are obtained as follows.

Method I	75	62	71	58	73
Method II	81	85	68	92	90
Method III	73	69	60	75	81

Determine at (a) 0.05 (b) 0.01 level whether there is a significant difference in the teaching methods.

Solution :

H_0 : $\mu_1 = \mu_2 = \mu_3$ i.e. there is no significant difference in teaching methods with regard to marks obtained.

H_1 : There is a significant difference in teaching methods.

Computation of test statistic F :

Coding of data has been done by subtracting 68 from each item. Let X_1 , X_2 and X_3 represent coded data for methods I, II and III respectively.

Methods : Coded data			Sum of squares		
X_1	X_2	X_3	X_1^2	X_2^2	X_3^2
7	13	5	49	169	25
-6	17	1	36	289	1
3	0	-8	9	0	64
-10	24	7	100	576	49
5	22	13	25	484	169
-1	76	18	219	1518	308

$$T = \sum X_1 + \sum X_2 + \sum X_3 = -1 + 76 + 18 = 93$$

$$\text{Correction factor (C.F)} = T^2/N = (93)^2/15 = 576.6$$

Sum of squares between methods (SSC)

$$= \frac{(\sum X_1)^2}{n_1} + \frac{(\sum X_2)^2}{n_2} + \frac{(\sum X_3)^2}{n_3} - \frac{T^2}{N}$$

$$= \frac{(-1)^2}{5} + \frac{(76)^2}{5} + \frac{(18)^2}{5} - 576.6 = 643.6$$

Total sum of squares (SST)

$$= \sum X_1^2 + \sum X_2^2 + \sum X_3^2 - T^2/N$$

$$= 219 + 1518 + 308 - 576.6 = 1468.4$$

Sum of squares within methods SSE = SST - SSC

$$= 1468.4 - 643.6 = 824.8$$

ANOVA Table

Source of variation	Sum of squares S.S.	Degrees of freedom d.f.	Mean sum of squares M.S.	F - ratio
Between Methods	643.6	$3 - 1 = 2$	321.8	$\frac{321.8}{68.73}$
Within Methods	824.8	$15 - 3 = 12$	68.73	$= 4.68$
Total	1468.4	$15 - 1 = 14$		

(a) *Tabulated value* of $F_{0.05}$ for $\nu_1 = 2$ and $\nu_2 = 12$ is 3.89. Since computed value of F is greater than its tabulated value, H_0 is rejected i.e. there is a significant difference in teaching methods.

(b) *Tabulated value* of $F_{0.01}$ for $\nu_1 = 2$ and $\nu_2 = 12$ is 6.93. Since computed value of F is less than its tabulated value, H_0 is accepted i.e. there is no significant difference in teaching methods with regard to marks obtained.

Thus we conclude that the teaching methods are different at 5% level and they are not different at 1% level of significance.

Example 6.

Final grades given by three professors in an advanced course are as follows :

A	:	63	45	73	77	72		
B	:	67	45	76	80	70	70	
C	:	97	97	87	87	84	74	64

Does the difference in sample means appear to be due to chance variation or is there significant evidence to indicate that not all population means are the same ? Use 5% level of significance.

Solution :

Null hypothesis (H_0) : $\mu_1 = \mu_2 = \mu_3$ i.e. the difference in sample mean is due to chance variation.

Alternative Hypothesis (H_1) : $\mu_1 \neq \mu_2 \neq \mu_3$

where suffices 1, 2, and 3 indicate professors A, B and C respectively.

Computation of test statistic F :

To simplify we use coded data by subtracting 70 from each item.

Professors : Coded data			Sum of squares		
X_1	X_2	X_3	X_1^2	X_2^2	X_3^2
-7	-3	27	49	9	729
-25	-25	27	625	625	729
3	6	17	9	36	289
7	10	17	49	100	289
2	0	14	4	0	196
	0	4		0	16
		-6			36
-20	12	100	736	770	2284

Sum of all items, $T = \sum X_1 + \sum X_2 + \sum X_3$

$$= -20 - 12 + 100 = 68$$

$$\text{Correction factor (C.F)} = \frac{T^2}{N} = \frac{(68)^2}{18} = 256.89$$

$$\begin{aligned} \text{Total sum of squares, SST} &= \sum X_1^2 + \sum X_2^2 + \sum X_3^2 - T^2/N \\ &= 736 + 770 + 2284 - 256.89 \\ &= 3533.11 \end{aligned}$$

Sum of squares between the samples (professors)

$$\begin{aligned} \text{SSC} &= \frac{(\sum X_1)^2}{n_1} + \frac{(\sum X_2)^2}{n_2} + \frac{(\sum X_3)^2}{n_3} - \frac{T^2}{N} \\ &= \frac{(-20)^2}{5} + \frac{(-12)^2}{6} + \frac{(100)^2}{7} - 256.89 = 1275.68 \end{aligned}$$

Sum of squares within samples (professors),

$$\begin{aligned} \text{SSE} &= \text{SST} - \text{SSC} \\ &= 3533.11 - 1275.68 = 2257.43 \end{aligned}$$

ANOVA Table

Source of variation	Sum of squares S.S.	Degrees of freedom d.f.	Mean sum of squares M.S.	F - ratio
Between Samples	1275.68	$3 - 1 = 2$	$1275.68/2 = 637.84$	$\frac{637.84}{150.5} = 4.24$
Within Samples	2257.43	$18 - 3 = 15$	$2257.43/15 = 150.5$	
Total	3533.11	$18 - 1 = 17$		

The tabulated value of $F_{0.05}$ for $v_1 = 2$ and $v_2 = 15$ is 3.68.

Decision : Since computed value of $F (= 4.24) > F_{0.05} (= 3.68)$, H_0 is rejected i.e. the difference of sample means is not due to chance variation.

Example 7.

A manufacturer of machine parts is considering buying one of 5 machines in the market. The following is the daily output on five randomly selected days for each machine.

<u>Machine 1</u>	<u>Machine 2</u>	<u>Machine 3</u>	<u>Machine 4</u>	<u>Machine 5</u>
72	62	68	64	72
56	70	72	72	62
68	66	74	68	68
65	64	70	68	64
60	78	66	58	70

Do the five machines have equal output rate ?

Solution :

H_0 : $\mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$ i.e. there is no difference between the machines with regard to outputs.

H_1 : The five machines have not equal output rate.

Computation of test statistic F :

Coding of data is done by subtracting 68 from each item. Let X_1, X_2, X_3, X_4 and X_5 denote the coded output data of machines 1, 2, 3, 4 and 5 respectively.

Machine 1		Machine 2		Machine 3		Machine 4		Machine 5	
X_1	X_1^2	X_2	X_2^2	X_3	X_3^2	X_4	X_4^2	X_5	X_5^2
4	16	-6	36	0	0	-4	16	4	16
-12	144	2	4	4	16	4	16	-6	36
0	0	-2	4	6	36	0	0	0	0
-3	9	-4	16	2	4	0	0	-4	16
-8	64	10	100	-2	4	-10	100	2	4
-19	233	0	160	10	60	-10	132	-4	72

$$T = \sum X_1 + \sum X_2 + \sum X_3 + \sum X_4 + \sum X_5$$

$$= -19 + 0 + 10 - 10 - 4 = -23$$

$$\text{Correction factor (CF)} = T^2/N = (-23)^2/25 = 21.16$$

$$SST = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 + \sum X_5^2 - T^2/N$$

$$= 233 + 160 + 60 + 132 + 72 - 21.16 = 635.84$$

$$SSC = \frac{(\sum X_1)^2}{n_1} + \frac{(\sum X_2)^2}{n_2} + \frac{(\sum X_3)^2}{n_3} + \frac{(\sum X_4)^2}{n_4} + \frac{(\sum X_5)^2}{n_5} - \frac{T^2}{N}$$

$$= \frac{(-19)^2}{5} + \frac{0^2}{5} + \frac{10^2}{5} + \frac{(-10)^2}{5} + \frac{(-4)^2}{5} - 21.16 = 94.24$$

$$\text{Sum of squares within samples, SSE} = SST - SSC$$

$$= 635.84 - 94.24 = 541.60$$

ANOVA Table

Source of variation	Sum of squares S.S.	Degrees of freedom df	Mean sum of squares M.S.	F - ratio
Within Machines	541.60	$25 - 5 = 20$	27.08	$\frac{27.08}{23.56}$
Between Machines	94.24	$5 - 1 = 4$	23.56	$= 1.15$
Total	635.84	$25 - 1 = 24$		

Tabulated value of $F_{0.05}$ for $v_1 = 20$ and $v_2 = 4$ is 5.80.

Decision : Since the computed value of $F (= 1.15)$ is less than tabulated

$F_{0.05} (= 5.80)$, H_0 is accepted i.e. there is no difference between the machine with regard to output. Thus we are 95% confident that the five machines have almost equal output rate.

8.3.2 ANOVA in Two-way Classification

In one-way classification we study the effect of one factor on various sample groups. In two way classification we study the effect of two factors i.e. the data are classified, according to the two different factors. For example, fertilisers may be tried on different soil texture. Thus we can classify the yield data with fertilisers in the column and the different types of textures in the rows. But there may be sampling variations besides these two factors which is called *residual variations*.

$$\text{We have, } SST = SSC + SSR + SSE$$

where SST = total sum of squares of variation

SSC = sum of squares of variations in columns

SSR = sum of squares of variations in rows

SSE = sum of squares of variations as residual due to errors.

The total number of degrees of freedom $= k.r - 1 = c.r - 1 = N - 1$, where k and r refer to the number of columns and rows respectively.

$$d.f. \text{ between columns} = k - 1$$

$$d.f. \text{ between rows} = r - 1$$

$$d.f. \text{ between residuals} = (k - 1)(r - 1)$$

ANOVA Table

Sources of variation	Sum of squares	Degrees of freedom	Mean sum of squares	F - ratio
Between cols	SSC	$k - 1$	$MSC = \frac{SSC}{k - 1}$	$F_c = \frac{MSC}{MSE}$
Between rows	SSR	$r - 1$	$MSR = \frac{SSR}{r - 1}$	$F_r = \frac{MSR}{MSE}$
Residual	SSE	$(k-1)(r-1)$	$MSE = \frac{SSE}{(k-1)(r-1)}$	
Total	SST	$N - 1$		

The steps 1, 3 and 4 are the same as those of Direct Method.

Example 8.

You are given the following data relating to the number of Table Fans sold by salesmen in three months, April, May and June.

Month	Salesmen			
	Ram	Mohan	Vijay	Krishna
April	50	40	48	39
May	46	48	50	45
June	39	44	40	39

Is there a significant difference in the sales made (i) by four salesmen and (ii) during different months ?

Solution :

Formulation of hypothesis

Null hypothesis (H_0) :

- (i) There is no difference in the sales made by 4 salesmen
- (ii) There is no difference in the sales made during three different months.

Computation of test statistic F :

To simplify the computation, the coding of data is done by subtracting 45 from each item. Let X_1 , X_2 , X_3 and X_4 represent the coded data of salesmen Ram, Mohan, Vijaya and Krishna respectively.

Month	Salesmen					Salesmen				
	X ₁	X ₂	X ₃	X ₄	ΣX _r	X ₁ ²	X ₂ ²	X ₃ ²	X ₄ ²	ΣX _r ²
April	5	-5	3	-6	-3	25	25	9	36	95
May	1	3	5	0	9	1	9	25	0	35
June	-6	-1	-5	-6	-18	36	1	25	36	98
ΣX _c	0	-3	3	-12	T =-12	62	35	59	72	228

$$\text{Correction factor C.F.} = \frac{T^2}{N} = \frac{(-12)^2}{12} = \frac{144}{12} = 12 \text{ (since } N = \text{c.r.)}$$

Sum of squares between salesmen

$$\text{SSC} = \frac{\sum (\Sigma X_c)^2}{n_r} - \frac{T^2}{N} = \left[\frac{0^2 + (-3)^2 + 3^2 + (-12)^2}{3} \right] - 12 = 42$$

Sum of squares between months,

$$\text{SSR} = \frac{\sum (\Sigma X_r)^2}{n_c} - \frac{T^2}{N} = \left[\frac{(-3)^2 + 9^2 + (-18)^2}{4} \right] - 12 = 91.5$$

$$\text{SST} = \sum (X_1^2 + X_2^2 + X_3^2 + X_4^2) - \text{C.F.} = 228 - 12 = 216$$

$$\begin{aligned} \text{Residual or Error, SSE} &= \text{SST} - (\text{SSC} + \text{SSR}) \\ &= 216 - (42 + 91.5) = 82.5 \end{aligned}$$

ANOVA Table

Source of variation	S.S.	d.f	M.S.	F - ratio
Between Salesmen	42	4 - 1 = 3	42/3 = 14	14/13.75 = 1.018
Between Months	91.5	3 - 1 = 2	91.5/2 = 45.75	45.75/13.75 = 3.33
Residual	82.5	(4-1) (3-1) = 6	82.5/6 = 13.75	
Total	216.0	11		

- (i) The tabulated value of $F_{0.05}$ (for $v_1 = 3$, $v_2 = 6$) = 4.76

Decision : Since $F (= 1.018) < F_{0.05}$, H_0 is accepted i.e. there is no significant difference in the sales made by 4 salesmen.

- (ii) The tabulated value of $F_{0.05}$ (for $v_1 = 2$, $v_2 = 6$) = 5.14

Decision : Since $F (= 3.33) < F_{0.05}$, H_0 is accepted i.e. there is no significant difference in the sales made during three different months.

Example 9.

Articles manufactured by a company are produced by 3 operators using 3 different machines. The manufacturer wishes to determine whether there is a difference (a) between operators (b) between machines at 5% level of significance. An experiment is performed to determine the number of articles per day produced by each operator using each machine, the results are given as follows:

	Operator 1	Operator 2	Operator 3
Machine A	23	27	24
Machine B	34	30	28
Machine C	28	25	27

Solution :

H_0 : There is no significant difference between (a) operators (b) machines

H_1 : There is significant difference between (a) operators (b) machines

Computation of test statistic F

To simplify the computation, the coding of data is done by subtracting 27 from each item.

Coded Operator data:					Coded data : Sum of squares			
Machine	X ₁	X ₂	X ₃	ΣX _r	X ₁ ²	X ₂ ²	X ₃ ²	ΣX _r ²
A	-4	0	-3	-7	16	0	9	25
B	7	3	1	11	49	9	1	59
C	1	-2	0	-1	1	4	0	5
ΣX _c	4	1	-2	T = 3	66	13	10	89

$$\text{Correction factor C.F.} = \frac{T^2}{N} = \frac{3^2}{9} = 1, \quad [N = \text{c.r}]$$

Sum of squares between operators

$$\text{SSC} = \frac{\sum (\sum X_c)^2}{n_r} - \frac{T^2}{N} = \frac{4^2 + 1^2 + (-2)^2}{3} - 1 = 6$$

Sum of squares between machines :

$$\begin{aligned} \text{SSR} &= \frac{\sum (\sum X_r)^2}{n_c} - \frac{T^2}{N} \\ &= \frac{(-7)^2 + 11^2 + (-1)^2}{3} - 1 = 56 \end{aligned}$$

$$\text{SST} = \sum (X_1^2 + X_2^2 + X_3^2) - T^2/N = 89 - 1 = 88$$

$$\text{Residual or Error, SSE} = \text{SST} - (\text{SSC} + \text{SSR}) = 88 - (6 + 56) = 26$$

ANOVA Table

Source of variation	Sum of squares S.S.	Degrees of freedom d.f.	Mean sum of squares M.S.	F - ratio
Between Operators	6	3 - 1 = 2	6/2 = 3	3/6.5 = 0.462
Between Machines	56	3 - 1 = 2	56/2 = 28	28/6.5 = 4.30
Residual	26	2 x 2 = 4	26/4 = 6.5	
Total	88	8		

Decision :

- (a) *Tabulated value* of $F_{0.05}$ for $\nu_1 = 2$ and $\nu_2 = 4$ is 6.94. Since the computed value of F (0.462) for different operation (columns) is less than the tabulated value (= 6.94), H_0 is accepted i.e. there is no significant difference in operators.
- (b) Since the computed value of F (= 4.3) for different machines (row) is less than tabulated value (= 6.94), H_0 is accepted i.e. there is no significant difference in machines.

Example 10.

Three varieties of wheat W_1 , W_2 and W_3 are treated with four different fertilisers f_1 , f_2 , f_3 and f_4 . The yields of wheat per acre are as under:

Fertiliser	Varieties of wheat		
	W_1	W_2	W_3
f_1	55	72	47
f_2	64	66	53
f_3	58	57	74
f_4	59	57	58

Set up a table for the analysis of variance and work out the F-ratios in respect of the above. Are the F - ratios significant ?

Solution :

H_0 : (i) There is no difference in the average yield of wheat in the three varieties.

(ii) There is no difference in the average yield of wheat when four different fertilisers are used.

Computation of test statistic F :

Coding of data may be done by subtracting 58 from each item.

Coded data: Varieties					Sum of squares			
Machine	X ₁	X ₂	X ₃	ΣX _r	X ₁ ²	X ₂ ²	X ₃ ²	(ΣX _r) ²
f_1	-3	14	-11	0	9	196	121	326
f_2	6	8	-5	9	36	64	25	125
f_3	0	-1	16	15	0	1	256	257
f_4	1	-1	0	0	1	1	0	2
ΣX _c	4	20	0	T = 24	46	262	402	710

$$\text{Correction factor C.F.} = \frac{T^2}{N} = \frac{24^2}{12} = 48$$

Sum of squares between varieties

$$\text{SSC} = \frac{\sum (\sum X_c)^2}{n_r} - \frac{T^2}{N} = \frac{4^2 + 20^2 + 0^2}{4} - 48 = 104 - 48 = 56$$

Sum of squares between fertilisers,

$$\text{SSR} = \frac{\sum (\sum X_r)^2}{n_c} - \frac{T^2}{N} = \frac{0^2 + 9^2 + 15^2 + 0^2}{3} - 48 = 54$$

$$\text{SST} = \sum (X_1^2 + X_2^2 + X_3^2 + X_4^2) - T^2/N = 710 - 48 = 662$$

$$\text{Residual, SSE} = \text{SST} - (\text{SSC} + \text{SSR}) = 662 - (56 + 54) = 552$$

ANOVA Table

Source	S.S.	d.f.	M.S.	F - ratio
Between Varieties	56	3 - 1 = 2	56/2 = 28	28/92 = 0.304
Between Fertilisers	54	4 - 1 = 3	54/3 = 18	18/92 = 0.196
Residual	552	2 x 3 = 6	552/6 = 92	
Total	662	11		

Tabulated value of $F_{0.05}$ for $v_1 = 2$ and $v_2 = 6$ is 5.14

Tabulated value of $F_{0.05}$ for $v_1 = 2$ and $v_2 = 6$ is 4.76

Decision :

- Since the computed value of F (= 0.304) is less than tabulated value (= 5.14), H_0 is accepted i.e. there is no difference in the average yield of wheat in the three varieties.
- Since the computed value of F (= 0.196) is less than tabulated value (= 4.76), H_0 is accepted i.e. there is no difference in the average yield of wheat when 4 different fertilisers are used.

Example 11.

Following table gives the yearly sales (in thousand rupees) of a firm in three territories by its four salesmen.

Territory	Salesmen			
	S_1	S_2	S_3	S_4
A	10	9	8	14
B	15	16	10	16
C	18	12	11	15

Set up the ANOVA table and state whether the difference between the sales effected by four salesmen and the difference between the sales in the three territories are significant.

Solution :

H_0 : (i) There is no difference between the sales effected by four salesmen.

(ii) There is no difference between the sales in the three territories.

Computation of test statistic F :

Coding of data may be done by subtracting 10 from each item.

Coded data Territory	Salesmen					Sum of squares				
	S_1	S_2	S_3	S_4	$\sum S_r$	S_1^2	S_2^2	S_3^2	S_4^2	$\sum S_r^2$
A	0	-1	-2	4	1	0	1	4	16	21
B	5	6	0	6	17	25	36	0	36	97
C	8	2	1	5	16	64	4	1	25	94
$\sum S_c$	13	7	-1	15	$T = 34$	89	41	5	77	212

$$\text{Correction factor C.F.} = \frac{T^2}{N} = \frac{(34)^2}{12} = 96.33 \quad [N = c.r]$$

Sum of squares between salesmen

$$\text{SSC} = \frac{\sum (\sum S_c)^2}{n_r} - \frac{T^2}{N} = \frac{13^2 + 7^2 + (-1)^2 + 15^2}{3} - 96.33 = 51.67$$

Sum of squares between territories,

$$\text{SSR} = \frac{\sum (\sum S_r)^2}{n_c} - \frac{T^2}{N} = \frac{1^2 + 17^2 + 16^2}{4} - 96.33 = 40.17$$

$$\text{SST} = \sum (S_1^2 + S_2^2 + S_3^2 + S_4^2) - T^2/N$$

$$= 212 - 96.33 = 115.67$$

$$\text{Residual, SSE} = \text{SST} - (\text{SSC} + \text{SSR})$$

$$= 115.67 - (51.67 + 40.17)$$

$$= 23.83$$

ANOVA Table

Source	S.S.	d.f.	M.S.	F - ratio
Between Salesmen	51.67	4-1 = 3	51.67/3 = 17.22	17.22/3.97 = 4.34
Between Tertiaries	40.17	3-1 = 2	40.17/2 = 20.08	20.08/3.97 = 5.06
Residual	23.83	3 x 2 = 6	23.83/6 = 3.97	
Total	115.67	11		

Tabulated value of $F_{0.05}$ for $\nu_1 = 3$ and $\nu_2 = 6$ is 4.76

Tabulated value of $F_{0.05}$ for $\nu_1 = 2$ and $\nu_2 = 6$ is 5.14

Decision : (i) Since the computed value of F (=4.34) is slightly less than tabulated value of $F_{0.05}$ (= 4.76), H_0 is accepted, i.e. there may be no difference between the sales effected by four salesmen.

(ii) Since the computed value of F (= 5.06) is slightly less than tabulated $F_{0.05}$ (=5.14), H_0 is accepted i.e. there may be no difference between the sales in the three tertiaries.

Example 12.

Four experiments determine the moisture content of sample of a powder, each man taking a sample from each of six consignments and their assessments are :

	Consignment					
Observer	1	2	3	4	5	6
1	9	10	9	10	11	11
2	12	11	9	11	10	10
3	11	10	10	12	11	10
4	12	13	11	14	12	10

Perform an analysis of variance of these data and discuss if there is any significant difference between consignments or between observers.

Solution :

H_0 : There is significant difference between (a) consignments (b) observers.

Coding of data may be done by subtracting 11 from each item.

Consignments							Total
Observers	1	2	3	4	5	6	
1	-2	-1	-2	-1	0	0	-6
2	1	0	-2	0	-1	-1	-3
3	0	-1	-1	1	0	-1	-2
4	1	2	0	3	1	-1	6
	0	0	-5	3	0	-3	T = -5

$$\text{Correction factor} = \frac{T^2}{N} = \frac{(-5)^2}{24} = 1.04$$

$$\text{SST} = (-2)^2 + (-1)^2 + (-2)^2 + (-1)^2 + (1)^2 + (-2)^2 + (-1)^2 + (-1)^2 + (-1)^2 + 1^2 + (-1)^2 + (1)^2 + 2^2 + 3^2 + 1^2 + (-1)^2 - \text{cf} = 37 - 1.04 = 35.96$$

Sum of squares between consignments :

$$\text{SSC} = \frac{0^2}{4} + \frac{0^2}{4} + \frac{(-5)^2}{4} + \frac{3^2}{4} + \frac{0^2}{4} + \frac{(-3)^2}{4} - \frac{T^2}{N}$$

$$= 10.75 - 1.04 = 9.71$$

Sum of squares between observers :

$$\text{SSR} = \frac{(-6)^2}{6} + \frac{(-3)^2}{6} + \frac{(-2)^2}{6} + \frac{6^2}{6} - \frac{T^2}{N}$$

$$= 14.17 - 1.04 = 13.13$$

Residual, SSE = SST - (SSC + SSR)

$$= 35.96 - (9.71 + 13.13) = 13.12$$

ANOVA Table

Source of variation	Sum of squares S.S.	Degrees of freedom d.f.	Mean sum of squares M.S.	F - ratio
Between consignments	9.71	5	1.942	1.942/0.875 = 2.22
Between observers	13.13	3	4.377	4.377/0.875 = 5.0
Residual	13.12	15	0.875	
Total	35.96	23		

Table value of $F_{0.05}$ (5, 15) is 2.90

Table value of $F_{0.05}(3, 15)$ is 3.29

Decision : (a) Since the computed $F (= 2.22)$ is less than tabulated $F_{0.05} (= 2.90)$, H_0 is accepted i.e. there is no significance difference between consignments.

(b) Since the computed $F (= 5)$ is greater than tabulated value $F_{0.05} (= 3.29)$, H_0 is rejected i.e. the difference between observers are significant.

Example 13.

In a sample of 8 observations, the sum of squared deviations of item from the mean was 94.5. In another sample of 10 observations, the value was found to be 101.7. Test whether the difference between the two population variances is significant at 5% level.

You are given that at 5% level, critical value of F , for $v_1 = 7$ and $v_2 = 9$ is 3.29 and for $v_1 = 8$ and $v_2 = 10$ its value is 3.07.

Solution :

$H_0 : \sigma_1^2 = \sigma_2^2$ i.e. there is no significant difference between the two population variances.

$$H_1 : \sigma_1^2 \neq \sigma_2^2$$

Computation test statistic F :

We are given, $n_1 = 8$, $\sum (X_1 - \bar{X}_1) = 94.5$

$n_2 = 10$, $\sum (X_2 - \bar{X}_2) = 101.7$

$$\therefore S_1^2 = \frac{1}{n_1 - 1} \sum (X_1 - \bar{X}_1)^2 = \frac{1}{8 - 1} \times 94.5 = 13.5$$

$$S_2^2 = \frac{1}{n_2 - 1} \sum (X_2 - \bar{X}_2)^2 = \frac{1}{10 - 1} \times 101.7 = 11.3$$

Since $S_2^2 < S_1^2$, the test statistic under H_0 is :

$$F = \frac{S_1^2}{S_2^2} = \frac{13.5}{11.3} = 1.195$$

Level of significance : $\alpha = 0.05$

d.f. for larger estimate of variance = $v_1 = n_1 - 1 = 7$

d.f. for smaller estimate of variance = $v_2 = n_2 - 1 = 9$

Critical value of F :

Table value of F at 5% level of significance for $v = 7$ and $v_2 = 9$ is 3.29

Decision : Since the computed value of F is less than its table value at 5% level of significance, H_0 is accepted i.e. there is no significance difference between the two population variances.

Example 14.

The labour productivity index of Nepal are recorded as below:

Source	Year		
	1985	1990	1995
Agriculture	100	125	138
Manufacturing	100	60	53
Community and social service	100	89	80

Does the labour productivity index vary due to the (1) difference in sector (2) difference in time period ?

Solution :

Formulation of hypothesis

H_0 : (a) The labour productivity does not vary due to the difference in sectors

(b) The labour productivity does not vary due to the difference in time period.

Computation of F

Coding of data is done by subtracting 100 from each item

Coded data source				$\sum S_r$	Sum of squares			$\sum S_r^2$ Total
Year	S_1	S_2	S_3		S_1^2	S_2^2	S_3^2	
1985	0	25	38	63	0	625	1444	2069
1990	0	-40	-47	-87	0	1600	2209	3809
1995	0	-11	-20	-31	0	121	400	521
Total	0	-27	-29	-55 =T	0	2346	4053	6399

$$\text{Correction factor} = \frac{T^2}{N} = \frac{(-55)^2}{9} = 336.11$$

Sum of squares between sources is given by

$$SSC = \frac{\sum (\sum S_c)^2}{n_r} - \frac{T^2}{N} \\ = \frac{O^2 + (-27)^2 + (-29)^2}{3} - 336.11 = 187.22$$

Sum of squares between time periods

$$SSR = \frac{\sum (\sum S_r)^2}{n_c} - \frac{T^2}{N} \\ = \frac{63^2 + (-87)^2 + (-31)^2}{3} - 336.11 = 3830.22$$

$$SST = \sum (S_1^2 + S_2^2 + S_3^2 + S_4^2) - \frac{T^2}{N}$$

$$= 6399 - 336.11 = 6062.89$$

$$\therefore SSE = SST - (SSC + SSR)$$

$$= 6062.89 - (187.22 + 3830.22)$$

$$= 2045.45$$

ANOVA Table

Source	SS	d.f.	M.S.	F-ratio
Between sources	187.22	3-1=2	93.61	0.183
Between periods	3830.22	3-1=2	1915.11	3.745
Residual	2045.45	2×2=4	511.36	
Total	6062.89	8		

(a) Table value of $F_{0.05}$ for $v_1 = 2$ and $v_2 = 4$ is 6.94. Since the calculated value of $F (=0.183)$ is less than its table value ($=6.94$) if we accept H_0 i.e. the labour productivity does not vary due the difference in sectors.

(b) Table value of $F_{0.05}$ for $v_1 = 2$ and $v_2 = 4$ is 6.94. Since the computed value of F is less than its table value, H_0 is accepted.

Example 15.

A medical experiment was made to test the additional hours of sleep due to three drugs A, B, C tried on one patient each from different age groups. Examine whether age has not any significant effect as the gain in sleep. Further examine whether the three drugs are similar in their effects or not. It is given that at 5% level of significance $F_{(2,6)} = 5.14$,

$$F_{(3,6)} = 4.76.$$

Drug Age group	30 – 40	40 – 50	50 – 60	60 – 70	Total
A	2.0	1.2	1.0	0.3	4.5
B	1.1	0.8	0.0	-0.1	1.8
C	1.5	1.3	0.9	0.1	3.8
Total	4.6	3.3	1.9	0.3	10.1

Solution :

H_0 : There is no significant difference between

(i) the drugs (ii) age groups with regard to gain in sleep.

H_1 : There is significant difference between

(i) the drugs (ii) age group with regard to gain in sleep.

Computation of F

Drug				Total	Sum of Squares			Total
Age Group	X_1	X_2	X_3	$\sum X_r$	X_1^2	X_2^2	X_3^2	$\sum X_r^2$
30-40	2	1.1	1.5	4.6	4	1.21	2.25	7.46
40-50	1.2	0.8	1.3	3.3	1.44	0.64	1.69	3.77
50-60	1.0	0.0	0.9	1.9	1.00	0	0.81	1.81
60-70	0.3	-0.1	0.1	0.3	0.09	0.01	0.01	0.11
$\sum X_c$	4.5	1.8	3.8	10.1	6.53	1.86	4.76	13.15

$$\text{Correction factor C.F.} = \frac{T^2}{N} = \frac{(10.1)^2}{12} = 8.5$$

Sum of squares between drugs :

$$SSC = \frac{\sum (\sum X_c)^2}{n_r} - \frac{T^2}{N} = \frac{(4.5)^2 + (1.8)^2 + (3.8)^2}{4} - 8.5 \\ = \frac{20.25 + 3.24 + 14.44}{4} - 8.5 = \frac{37.93}{4} - 8.5 = 0.9825$$

Sum of squares between age group :

$$SSR = \frac{\sum (\sum X_r)^2}{n_c} - \frac{T^2}{N} \\ = \frac{(4.6)^2 + (3.3)^2 + (1.9)^2 + (0.3)^2}{3} - 8.5$$

$$SST = \sum (X_1^2 + X_2^2 + X_3^2 + X_4^2) - \frac{T^2}{N} = 13.15 - 8.5 = 4.65$$

$$SSE = SST - (SSC + SSR) = 4.65 - (0.9825 + 3.42) = 0.2475$$

ANOVA Table

Source of variation	S.S.	d.f.	M.S.	F-ratio
Between drugs	0.9825	3-1=2	0.49125	$\frac{0.49125}{0.04125} = 11.91$
Between age-group	3.42	4-1 = 3	1.14	$\frac{1.14}{0.04125} = 27.64$
Residual	0.2475	2 x 3 = 6	0.04125	
	4.65	12-1=11		

Tabulated value of $F_{0.05}$ for $v_1 = 2$ and $v_2 = 6$ is 5.14.

Tabulated value of $F_{0.05}$ for $v_1 = 3$ and $v_2 = 6$ is 4.76

Decision : (i) Since the computed value of $F (=11.91)$ is greater than its table value $(=5.14)$, H_0 is rejected i.e. there is significant difference between the drugs with regard to gain in sleep.

(ii) Since the computed value of $F (= 27.64)$ is greater than its table value $(=4.76)$, H_0 is rejected i.e. there is significant difference between the age group with regard to gain in sleep.

Example 16.

In Kathmandu, a fast food chain feels it is gaining a bad reputation because it takes too long to serve the customers. Since the chain has 4 restaurants in this city, it is concerned with whether all 4 restaurants have the same average service time. One of the owners of the fast-food chain has decided to visit each of the stores and monitor the service time for 5 randomly selected customers. At his 4 noon-time visits, he records the following service times in minutes.

Restaurant I	3	4	5.5	3.5	4
Restaurant II	3	3.5	4.5	4	5.5
Restaurant III	2	3.5	5	6.5	6
Restaurant IV	3	4	5.5	2.5	3

- (a) Using a 5 percent significance level, do all the restaurants have the same mean service time ?
- (b) Based on his results, should the owner make any policy recommendations to any of the restaurant managers ?

Solution :

Formulation of Hypothesis :

$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$ i.e. all the restaurants have the same mean service time.

$H_1 : \mu_1 \neq \mu_2 \neq \mu_3 \neq \mu_4$

Computation of F

Restaurant				Sum of Squares			
X_1	X_2	X_3	X_4	X_1^2	X_2^2	X_3^2	X_4^2
3.0	3.0	2.0	3.0	9	9	4	9
4.0	3.5	3.5	4.0	16	12.25	12.25	16
5.5	4.5	5.0	5.5	30.25	20.25	25	30.25
3.5	4.0	6.5	2.5	12.25	16	42.25	6.25
4.0	5.5	6.0	3.0	16	30.25	36	9.00
20.0	20.5	23.0	18.0	83.50	87.75	119.5	70.50

Sum of all items, $T = \sum X_1 + \sum X_2 + \sum X_3 + \sum X_4 = 81.5$

$$\text{Correction factor, C.F.} = \frac{T^2}{N} = \frac{(81.5)^2}{20} = \frac{6642.25}{20}$$

$$= 332.1125 \approx 332$$

Total sum of squares, $SST = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2$

$$= 361.25$$

Sum of squares between the samples (restaurants) :

$$\begin{aligned} SSC &= \frac{(\sum X_1)^2}{n_1} + \frac{(\sum X_2)^2}{n_2} + \frac{(\sum X_3)^2}{n_3} + \frac{(\sum X_4)^2}{n_4} - \frac{T^2}{N} \\ &= \frac{(20)^2}{5} + \frac{(20.5)^2}{5} + \frac{(23)^2}{5} + \frac{(18)^2}{5} - 332 = 2.65 \end{aligned}$$

$$SSE = SST - SSC = 361.25 - 2.65 = 358.6$$

ANOVA Table

Variation	S.S.	d.f.	M.S.	F-ratio
Between samples	2.65	4-1 = 3	0.88	$\frac{22.41}{0.88}$
Within samples	358.6	20 - 4 = 16	22.41	= 25.47
Total		20-1 = 19		

The tabulated value of $F_{0.05}$ for $v_1 = 16$ and $v_2 = 3$ is 8.69

Decision : (a) Since $F > F_{0.05}(16, 3)$ H_0 is rejected. Thus all the four restaurants have not the same mean service time.

(b) Based on result (a), we can suggest that the owner should make policy recommendations to all the restaurant managers.

EXERCISE 8

1. In two samples of 10 children the increase in weight due to two different diets (D_1 and D_2) in the same period in lbs were :

D_1 : 8 5 7 8 3 2 7 6 5 7

D_2 : 3 7 5 6 5 4 4 5 3 6

Find whether their variance differ significantly.

2. The samples are drawn from two normal populations. From the following data test whether the two samples have the same variance at 5% level :

Sample I : 60 65 71 74 76 82 85 87

Sample II : 61 66 67 85 78 63 85 86 88 91

3. The following data present the yields in quintals of corn on ten subdivisions of equal area of two agricultural plots :

Plot I : 6.2 5.7 6.5 6.0 6.3 5.8 5.7 6.0 6.0 5.8

Plot II : 5.6 5.9 5.6 5.7 5.8 5.7 6.0 5.5 5.7 5.5

Test whether two sample taken from two random population have the same variance (5% point of F for $v_1 = 9$, $v_2 = 9$ is 3.8)

4. Two random samples drawn from two normal populations are:

Sample I : 28 30 32 33 33 29 34

Sample II : 29 30 30 24 27 29

Obtain the estimates of the variances of the populations and also test whether the populations have the same variances.

Given : $F_{0.05}(5, 6) = 4.39$; $F_{0.05}(6, 5) = 4.95$

5. (a) Two random samples drawn from the population are
 Sample I 10 16 22 25 23 20 24 26 27 29
 Sample II 17 20 25 25 27 28 30 31 30
 Test whether the two populations have the same variance.
 $(v_1 = 9, v_2 = 8, F_{0.05} = 3.39$ and $v_1 = 8, v_2 = 9, F_{0.05} = 3.23)$

- (b) Test whether two population have the same variance or not from the following :

Samples I

Sample II

$n_1 = 7$

$n_2 = 6$

$\sum (X_1 - \bar{X}_1)^2 = 320$

$\sum (X_2 - \bar{X}_2)^2 = 350$

6. The following are data on harvest (tonnes per acre) in 6 farms, using fertilisers A, B, C. Frame suitable hypothesis and test it :

A 85 84 79 84 88 84

B 83 94 96 85 87 90

C 74 76 73 81 76 75

7. The following table shows the lives in hours of randomly selected electric lamps from four batches.

Batch

Lives in hours

1 1600 1610 1650 1680 1700 1720 1800

2 1580 1640 1640 1700 1750

3 1460 1550 1600 1620 1640 1660 1740 1820

4 1510 1520 1530 1570 1600 1680

Perform an analysis of variance of these data and show that a significance test does not reject their homogeneity.

8. Kathmandu Traders Co. Ltd., wishes to test whether its three salesmen A, B and C tend to make sales of the same size or whether they differ in their selling ability as measured by the average size of their sales. During the last week there have been 14 sale calls – A made 5 calls, B made 4 calls and C made 5 calls. Following are the weekly sales record of the three salesmen:

A	B	C
Rs.	Rs.	Rs.
300	600	700
400	300	300
300	300	400
500	400	600
0	–	500

Perform the analysis of variance and draw your conclusions.

9. The three samples below have been obtained from normal populations with equal variances. Test the hypothesis at 5% level that the population means are equal.

Sample 1	Sample 2	Sample 3
8	7	12
10	5	9
7	10	13
14	9	12
11	9	14

(The table value of F at 5% level for $\nu_1=2$ and $\nu_2=12$ is 3.88)

10. A test was given to 5 students chosen at random from M.B.A. class of each of the three universities. Their scores were found as:

University	scores				
A	90	70	60	50	80
B	70	40	50	40	50
C	60	50	60	70	60

Perform analysis of variance and show if there is any significant difference between the scores of students in the three Universities.

(Given F at 5% level, = 3.44)

11. A special type of fertiliser was used in four agriculture fields A, B, C and D. Each field was divided into four beds and the fertiliser was applied over them. The respective yield of the beds of four fields are given below. Find whether the difference in mean yield of fields is significant or not ? (At 5%

level of significance the table value of F for $\nu_1 = 3$, and $\nu_2 = 12$ is 8.74)

Plot yield			
A	B	C	D
8	9	3	3
12	4	8	7
1	7	2	8
3	1	5	2

12. Four different manufacturing processes were tried at three different stations and the average measurement of a quality characteristic of the product by these processes obtained, as in the following table. Perform the analysis of variance of the data and list for the difference between the processes.

	Process			
Station	A	B	C	D
I	7	14	11	11
II	15	16	14	10
III	8	15	10	12

13. Following table gives the monthly sales (in thousand rupees) of a certain firm in three regions by its four salesmen.

	Salesmen			
Region	A	B	C	D
X	5	4	4	7
Y	7	8	5	4
Z	<u>9</u>	<u>6</u>	<u>6</u>	<u>7</u>
Total	21	18	15	18

Perform ANOVA.

14. A tea company appoints four salesmen A, B, C and D and observes their sales in three seasons - summer, winter and monsoon. The figures (in lakhs) are given in the following table :

Seasons	Salesmen				Total
	A	B	C	D	
Summer	36	36	21	35	128
Winter	28	29	31	32	120
Monsoon	<u>26</u>	<u>28</u>	<u>29</u>	<u>29</u>	<u>112</u>
Salesmen's Total	90	93	81	96	360

Perform ANOVA.

15. The following data represent the number of units of production per day turned out by 5 different workers using 4 different types of machines.

Workers	Machine type			
	A	B	C	D
1	44	38	47	36
2	46	40	52	43
3	34	36	44	32
4	33	38	46	33
5	38	42	49	39

- (a) Test whether the mean productivity is the same for different machine types.
- (b) Test whether five men differ with respect to mean productivity.
16. The following data represent the number of units of a commodity produced by 3 different workers using 3 different types of machines :

Machines/Workers	A	B	C
X	16	64	40
Y	56	72	56
Z	12	56	28

Test : (i) whether the mean productivity is the same for the different machine types, and (ii) whether the three workers differ with respect to mean productivity.

17. To study the performance of three detergents and three different water temperatures the following whiteness were obtained with specially designed equipment.

Water temperature	Detergent A	Detergent B	Detergent C
Cold water	57	55	67
Warm water	49	52	68
Hot water	54	46	58

Perform a two way analysis of variance using 5% level of significance (Given $F = 6.94$)

18. The following are the defective pieces produced by four operators working, in turn, on four different machines.

Machine	Operators			
	B/1	B/2	B/3	B/4
A/1	34	28	33	29
A/2	31	24	35	23
A/3	27	20	43	72
A/4	28	28	29	26

Perform analysis of variance at 0.05 level of significance to ascertain whether variability in production is due to variability in operators performance or variability in machines performance.

19. Set up ANOVA table for the following per hectare yield for three varieties of wheat each grown on four plots :

Per hectare yield (in hundred Kg.)

Plot of land	Variety of Wheat		
	A ₁	A ₂	A ₃
1	6	5	5
2	7	5	4
3	3	3	3
4	8	7	4

Also work out F-ratio.

20. Three training methods were compared to see if they led to greater productivity after training. Below are productivity measures for individuals trained by each method.

Method 1	36	26	31	20	34	25
Method 2	40	29	38	32	39	34
Method 3	32	18	23	21	33	27

At the 0.05 level of significance, do the three training methods lead to different levels of productivity ?